Unit III :: Conservation Laws

PHY01MX01R04: Mechanics and Properties of Matter

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Syllabus

Unit III: Conservation Laws

Laboratory and centre-of-mass frames of reference; principle of conservation of linear momentum; collisions in one and two dimensions – elastic and inelastic; variable-mass systems, rocket propulsion; principle of conservation of angular momentum; applications: planetary motion, alpha-particle scattering, effect on linear and angular speeds of a particle upon orbital contraction, shape of galaxies.

1 Frames of reference

1.1 Laboratory frame

1.2 Centre-of-mass frame

1.3 Conservation of linear momentum

2 Collisions

3 Variable mass systems

3.1 Rocket propulsion

4 Conservation of angular momentum

5 Applications of conservation laws

5.1 Planetary motion

5.2 Scattering of α-particles

5.3 Orbital contraction

5.4 Shape of galaxies